

Vector Equations

- A **vector** means ***an ordered list of numbers***. (More details about the term **vector** will be discussed in Chapter 4).
- A matrix with only **one column** is called a **column vector** or simply a **vector**

Examples of vectors with two entries

$$\mathbf{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} .2 \\ .3 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

The set of all vectors with two entries is $\mathbb{R}^2$

$\mathbb{R}$: real numbers

2: two entries

Vector Equations

Vector entries are ordered: $\begin{bmatrix} 4 \\ 7 \end{bmatrix}$ and $\begin{bmatrix} 7 \\ 4 \end{bmatrix}$ are NOT equal

Given two vectors $\mathbf{v}$ and $\mathbf{u}$ in $\mathbb{R}^2$:

Addition: $\mathbf{v} + \mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1+2 \\ -2+5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

Scaler multiple by real number c (scalar) : the vector $c\mathbf{u}$

if $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $c = 5$, then $c\mathbf{u} = 5\begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$

Example

Given $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$, find $4\mathbf{u}$, $(-3)\mathbf{v}$, and $4\mathbf{u} + (-3)\mathbf{v}$.

SOLUTION

$$4\mathbf{u} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}, \quad (-3)\mathbf{v} = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$$

and

$$4\mathbf{u} + (-3)\mathbf{v} = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$$

Geometric Descriptions of $\mathbb{R}^2$

- Consider a rectangular coordinate system in the plane.

geometric point (a, b) $\rightarrow$ column vector $\begin{bmatrix} a \\ b \end{bmatrix}$

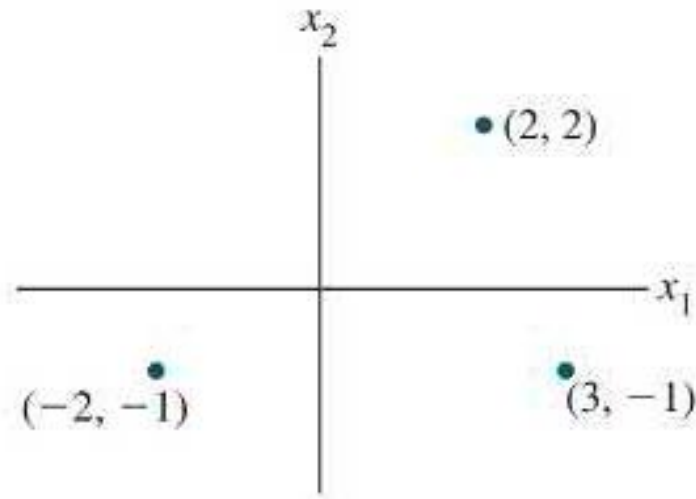


FIGURE 1 Vectors as points.

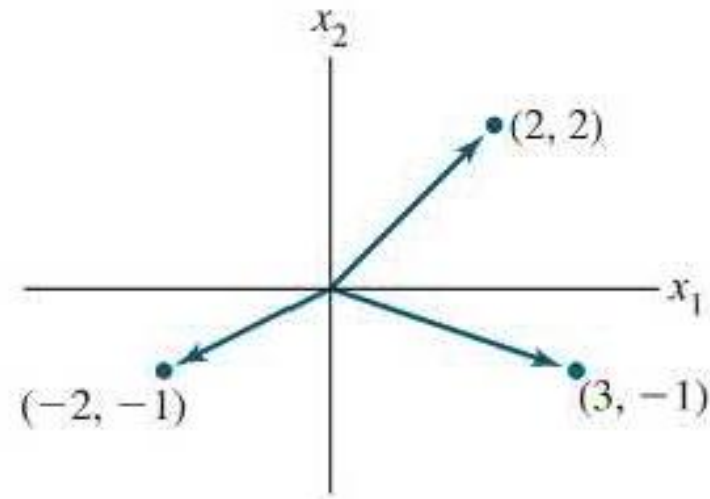


FIGURE 2 Vectors with arrows.

Geometric Descriptions of $\mathbb{R}^2$

Parallelogram Rule for Addition

If $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ are represented as points in the plane, then $\mathbf{u} + \mathbf{v}$ corresponds to the fourth vertex of the parallelogram whose other vertices are $\mathbf{u}$, $\mathbf{0}$, and $\mathbf{v}$. See Figure 3.

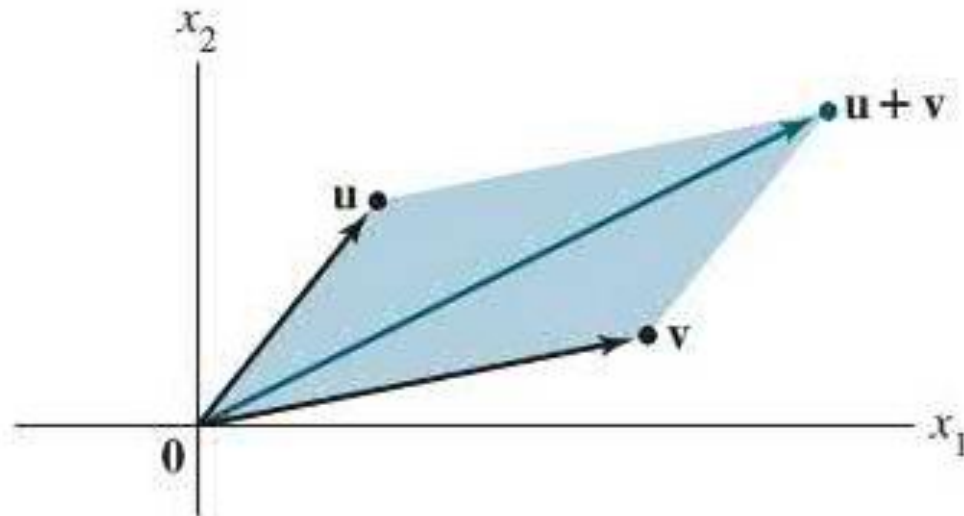
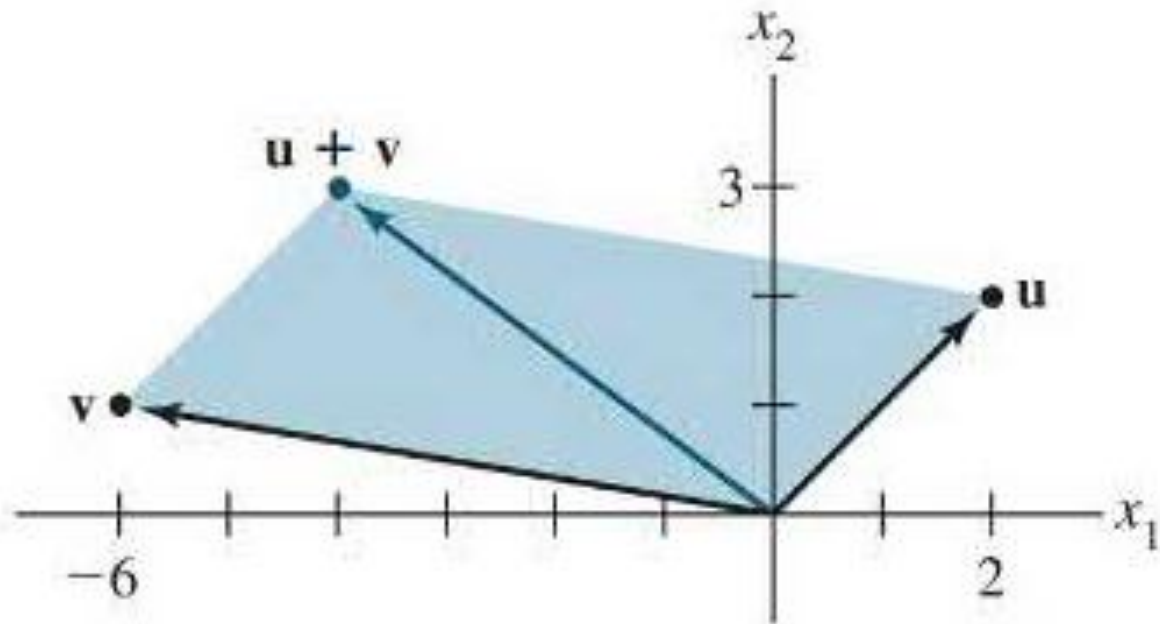


FIGURE 3 The parallelogram rule.

Example

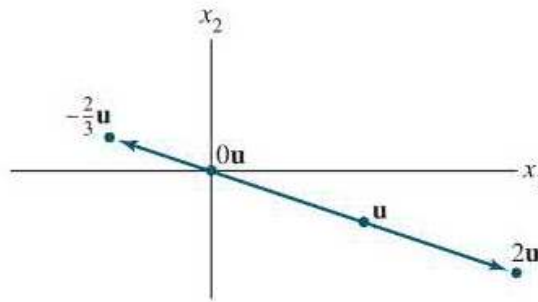
The vectors $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ are displayed



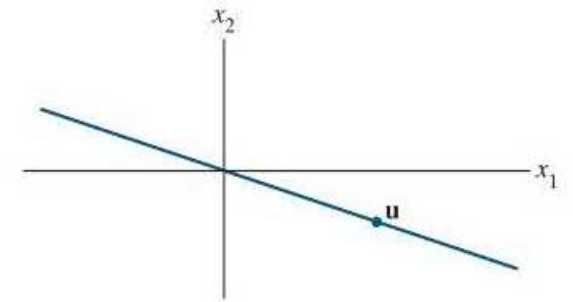
Example

Let $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. Display the vectors $\mathbf{u}$, $2\mathbf{u}$, and $-\frac{2}{3}\mathbf{u}$ on a graph.

$$2\mathbf{u} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}, \text{ and } -\frac{2}{3}\mathbf{u} = \begin{bmatrix} -2 \\ 2/3 \end{bmatrix}$$



Typical multiples of $\mathbf{u}$



The set of all multiples of $\mathbf{u}$

the length of the arrow for $c\mathbf{u}$ is $|c|$ * the length of the arrow for $\mathbf{u}$

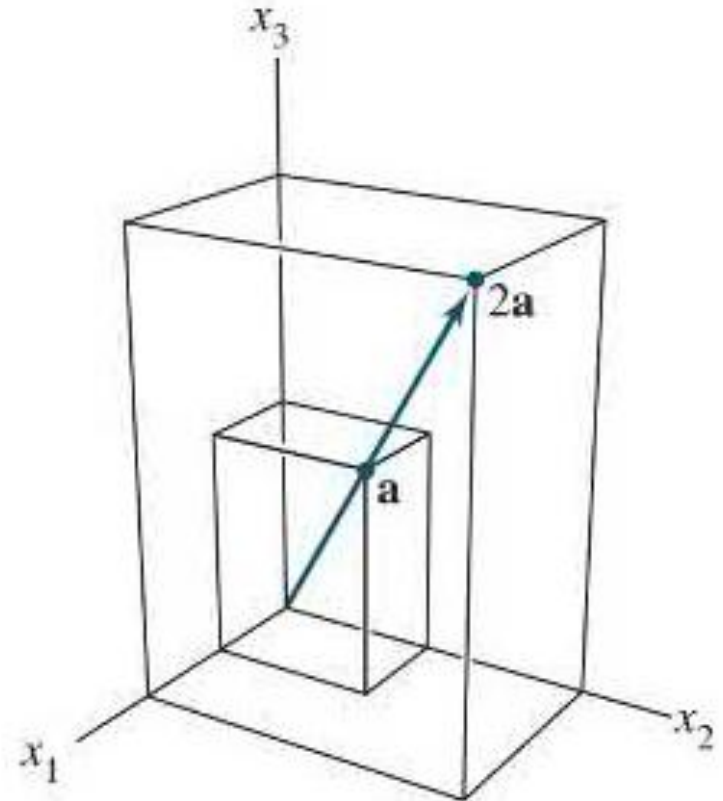
The length of the arrow for $\mathbf{u}$ with entries $\begin{bmatrix} a \\ b \end{bmatrix}$ is calculate $\sqrt{a^2 + b^2}$,
which

is the length of the line segment between the points $(0, 0)$ and (a, b)

Vectors in $\mathbb{R}^3$

- 3×1 column matrices with three entries
- Represented geometrically in the three-dimensional coordinate space

The vectors $\mathbf{a} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ and $2\mathbf{a}$ are displayed



Vectors in $\mathbb{R}^n$

- $n \times 1$ column matrices with n entries

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

- $\mathbb{R}^n$ is the collection of all ordered n -tuples

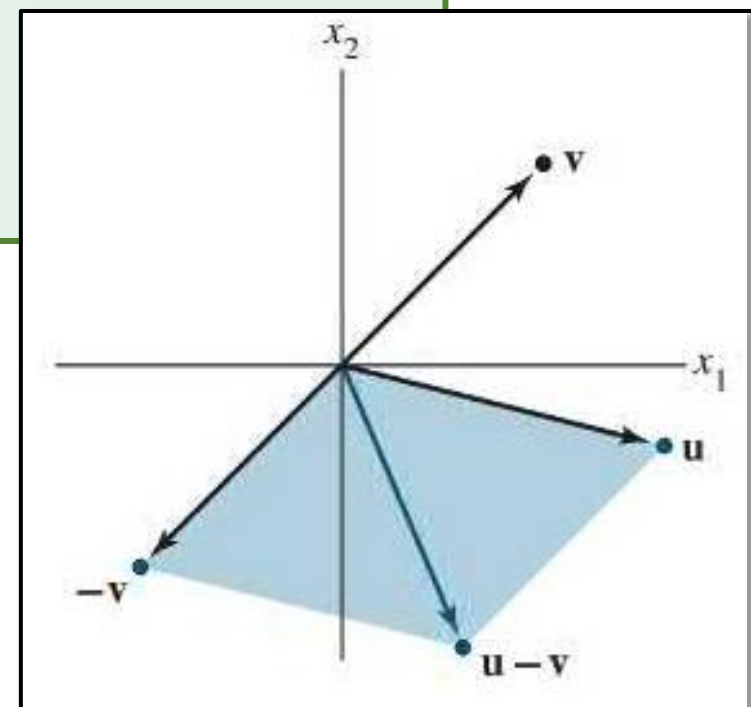
Algebraic Properties of $\mathbb{R}^n$

For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d :

- | | |
|---|--|
| (i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ | (v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ |
| (ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ | (vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ |
| (iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ | (vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$ |
| (iv) $\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$,
where $-\mathbf{u}$ denotes $(-1)\mathbf{u}$ | (viii) $1\mathbf{u} = \mathbf{u}$ |

The vector whose entries are all zero is called the **zero vector** and is denoted by $\mathbf{0}$

vector such as $\mathbf{u} + (-1)\mathbf{v}$ is often written as $\mathbf{u} - \mathbf{v}$.



Linear Combinations

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \dots, c_p$, the vector $\mathbf{y}$ defined by

$$\mathbf{y} = c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p$$

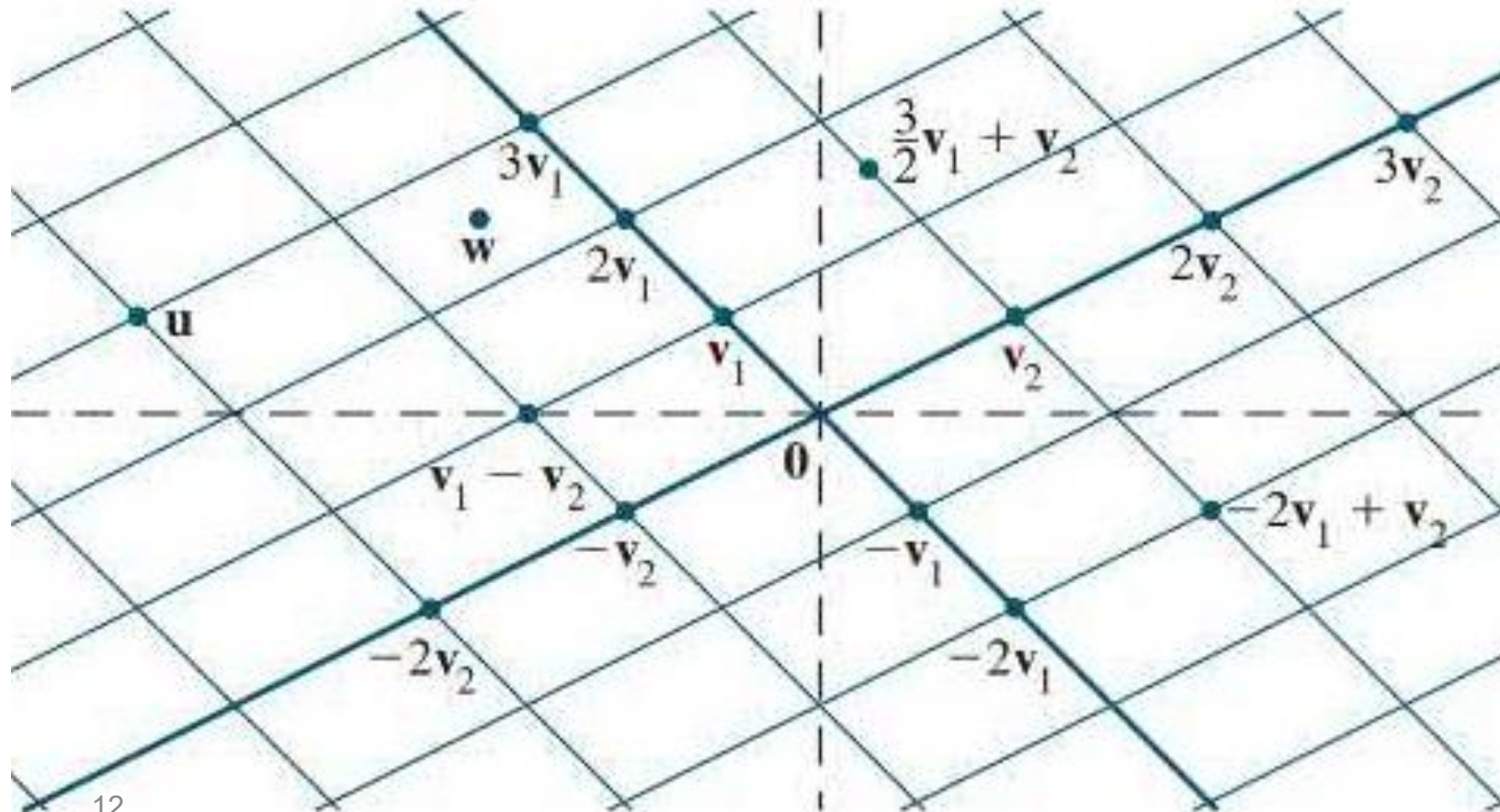
Any real
number
including zero

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_p$ with weights $c_1, \dots, c_p$.

some linear combinations of vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are

$$\sqrt{3} \mathbf{v}_1 + \mathbf{v}_2, \quad \frac{1}{2} \mathbf{v}_1 (= \frac{1}{2} \mathbf{v}_1 + 0 \mathbf{v}_2), \quad \text{and} \quad \mathbf{0} (= 0 \mathbf{v}_1 + 0 \mathbf{v}_2)$$

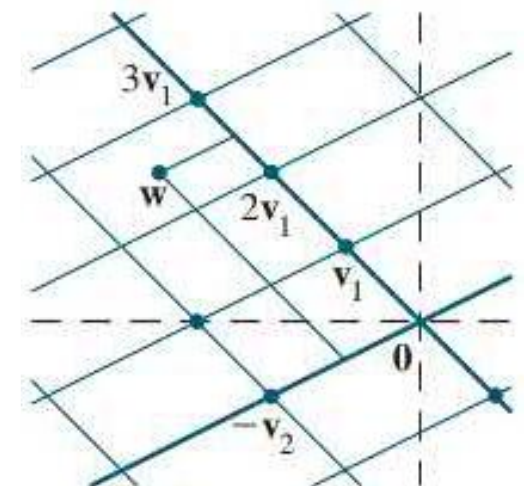
EXAMPLE 4 Figure 8 identifies selected linear combinations of $\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. (Note that sets of parallel grid lines are drawn through integer multiples of $\mathbf{v}_1$ and $\mathbf{v}_2$.) Estimate the linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$ that generate the vectors $\mathbf{u}$ and $\mathbf{w}$.



Solution:

$$\mathbf{u} = 3\mathbf{v}_1 - 2\mathbf{v}_2$$

$$\mathbf{w} = \frac{5}{2}\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_2$$



Linear Combinations

A vector equation

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{array} \right] \quad (5)$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$ if and only if there exists a solution to the linear system corresponding to the matrix (5).

Example

$$\text{Let } \mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Determine whether $\mathbf{b}$ can be generated as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Solution: If $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$, then

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b} \quad \rightarrow \quad \text{augmented matrix } [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{b}]$$

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_2 = R_2 + 2R_1 \\ R_3 = R_3 + 5R_1 \end{array}$$

$$R_2 = 1/9 R_2$$

$$\begin{array}{l} R_3 = R_3 - 16R_2 \\ R_1 = R_1 - 2R_2 \end{array}$$

Solution: $x_1 = 3$ and $x_2 = 2$

Since the system has a solution, then $\mathbf{b}$ can be generated as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$, and the combination is:

$$3\mathbf{a}_1 + 2\mathbf{a}_2 = \mathbf{b}$$

$$3 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Definition

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is denoted by $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $\mathbf{v}_1, \dots, \mathbf{v}_p$** . That is, $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the collection of all vectors that can be written in the form

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p$$

with $c_1, \dots, c_p$ scalars.

Note that $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ contains every scalar multiple of $\mathbf{v}_1$ (for example), since $c\mathbf{v}_1 = c\mathbf{v}_1 + 0\mathbf{v}_2 + \dots + 0\mathbf{v}_p$. In particular, the zero vector must be in $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

The following questions are equivalent:

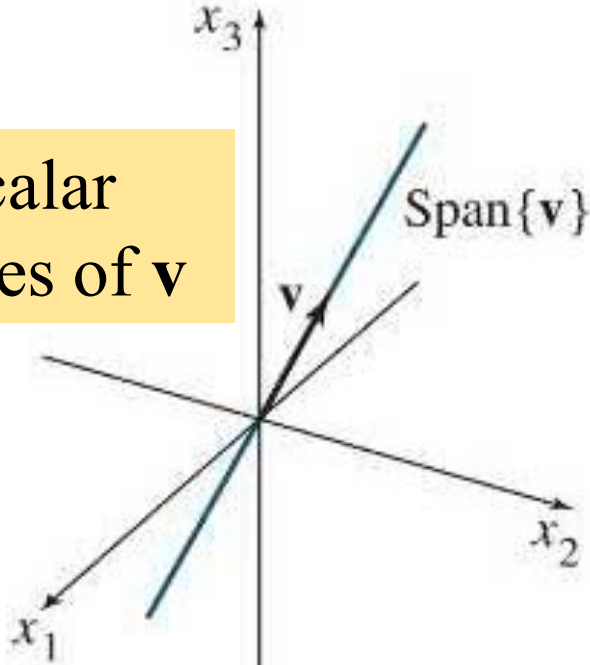
- Is vector **b** in $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$?
- Is there a solution for the equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{b}$
- Is there a solution for the augmented matrix $\begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_p & \mathbf{b} \end{bmatrix}$
- And some other questions explained next

A Geometric Description of $\text{Span}\{\mathbf{v}\}$ and $\text{Span}\{\mathbf{u}, \mathbf{v}\}$

Let $\mathbf{v}$ be a nonzero vector in $\mathbb{R}^3$.

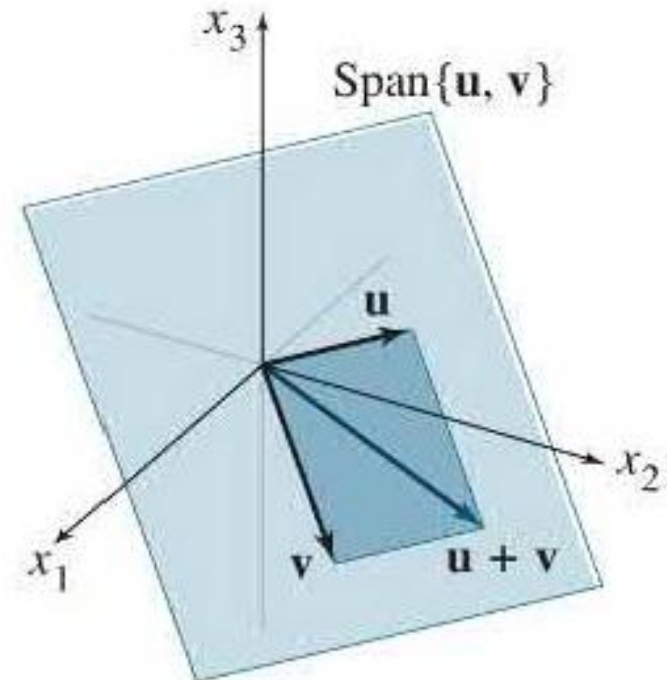
Then $\text{Span}\{\mathbf{v}\}$ is the set of all vectors that can be written as $c\mathbf{v}$ (i.e., the line passing through $\mathbf{v}$ and $\mathbf{0}$)

all scalar
multiples of $\mathbf{v}$



If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^3$, with $\mathbf{v}$ not a multiple of $\mathbf{u}$,

Then $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is the set of all vectors that can be written as $c_1\mathbf{u} + c_2\mathbf{v}$ (i.e., the plane containing $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{0}$)



EXAMPLE 6 Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$. Then

$\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$ is a plane through the origin in $\mathbb{R}^3$. Is $\mathbf{b}$ in that plane?

SOLUTION Does the equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b}$ have a solution? To answer this, row reduce the augmented matrix $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{b}]$:

$$\begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

The third equation is $0 = -2$, which shows that the system has no solution. The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b}$ has no solution, and so $\mathbf{b}$ is *not* in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$. ■

This is also another question that is equivalent to the following questions:

- Is vector $\mathbf{b}$ in $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$?
- Is there a solution for the equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{b}$?
- Is there a solution for the augmented matrix $[\mathbf{v}_1 \ \dots \ \mathbf{v}_p \ \mathbf{b}]$?

Linear Combinations in Applications

EXAMPLE 7 A company manufactures two products. For \$1.00 worth of product B, the company spends \$.45 on materials, \$.25 on labor, and \$.15 on overhead. For \$1.00 worth of product C, the company spends \$.40 on materials, \$.30 on labor, and \$.15 on overhead. Let

$$\mathbf{b} = \begin{bmatrix} .45 \\ .25 \\ .15 \end{bmatrix} \quad \text{and} \quad \mathbf{c} = \begin{bmatrix} .40 \\ .30 \\ .15 \end{bmatrix}$$

Then $\mathbf{b}$ and $\mathbf{c}$ represent the “costs per dollar of income” for the two products.

- a. What economic interpretation can be given to the vector $100\mathbf{b}$?

$$100\mathbf{b} = 100 \begin{bmatrix} .45 \\ .25 \\ .15 \end{bmatrix} = \begin{bmatrix} 45 \\ 25 \\ 15 \end{bmatrix}$$

The vector $100\mathbf{b}$ lists the various costs for producing \$100 worth of product B—namely \$45 for materials, \$25 for labor, and \$15 for overhead.

Linear Combinations in Applications

- b. Suppose the company wishes to manufacture x_1 dollars worth of product B and x_2 dollars worth of product C. Give a vector that describes the various costs the company will have (for materials, labor, and overhead).

The costs of manufacturing x_1 dollars worth of B are given by the vector $x_1\mathbf{b}$, and the costs of manufacturing x_2 dollars worth of C are given by $x_2\mathbf{c}$. Hence the total costs for both products are given by the vector $x_1\mathbf{b} + x_2\mathbf{c}$. 

The Matrix Equation $A\mathbf{x}=\mathbf{b}$

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the **product of A and $\mathbf{x}$** , denoted by $A\mathbf{x}$, is **the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights**; that is,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

Note that $A\mathbf{x}$ is defined only if the number of columns of A equals the number of entries in $\mathbf{x}$.

EXAMPLE 1

Matrix
multiplication

a. $\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ -15 \end{bmatrix} + \begin{bmatrix} -7 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

b. $\begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 8 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 32 \\ -20 \end{bmatrix} + \begin{bmatrix} -21 \\ 0 \\ 14 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix}$

Now, you know different ways to express a linear system

system of linear equations

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

e.g.,

$$\begin{aligned}x_1 + 2x_2 - x_3 &= 4 \\ -5x_2 + 3x_3 &= 1\end{aligned}$$

Vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

e.g., $x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

matrix equation

$$A\mathbf{x} = \mathbf{b}$$

e.g., $\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{b}$ is in $\mathbb{R}^m$, the matrix equation

$$A\mathbf{x} = \mathbf{b} \tag{4}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b} \tag{5}$$

which, in turn, has the same solution set as the system of linear equations whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix} \tag{6}$$

The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A .

EXAMPLE 3 Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible b_1, b_2, b_3 ?

SOLUTION Row reduce the augmented matrix for $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & b_3 + 3b_1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{bmatrix}$$

The third entry in column 4 equals $b_1 - \frac{1}{2}b_2 + b_3$. The equation $A\mathbf{x} = \mathbf{b}$ is *not* consistent for every $\mathbf{b}$ because some choices of $\mathbf{b}$ can make $b_1 - \frac{1}{2}b_2 + b_3$ nonzero. ■

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{array} \right]$$

Not all the rows of the **coefficient matrix** have pivots (in this case the last row has no pivot in it). Therefore, there is a possibility to see the following row (false condition) in **augmented matrix**:

$$[0 \ 0 \ 0 \ m] \text{ where } m \text{ is nonzero}$$

Which implies that the system is **inconsistent**.

In contrast, if we have a pivot in ALL rows of the **coefficient matrix**, then the system is **consistent** because it is impossible to see the false condition row in the **augmented matrix**.